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# Uncovering the role of juvenile hormone in ovary development and egg laying in bumble bees

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ABSTRACT

Juvenile hormone (JH) regulates developmental and physiological processes in insects. In bumble bees, the hormone acts as a gonadotropin that mediates ovary development, but the exact physiological pathways involved in ovary activation and subsequent egg laying are poorly understood. In this study, we examine how queen hibernation state, caste, and species impact the gonadotropic effect of JH in bumble bee queens through methoprene (JH analogue) application. We extend previous research by assessing queen egg laying and colony initiation, alongside ovary development. Furthermore, we compared sensitivity of workers of both species to the juvenile hormone's gonadotropic effect. In both bumble bee species, the ovaries of hibernated queens were developed five to six days after breaking diapause, regardless of methoprene treatment. By contrast, methoprene did have a stimulatory effect on ovary development in non-hibernated queens. The dose needed to obtain this effect was higher in *B. impatiens*. Methoprene did not have gonadotropic effects in callow workers of both species. These results indicate that the physiological effect of exogenous methoprene application varies according to species, caste and hibernation status. Interestingly, despite gonadotropic effects in non-hibernated queens, oviposition was not accelerated by JH. This suggests that JH alone is insufficient to induce egg laying and that an additional stimulus, which is naturally present in hibernated queens, is required. Consequently, our findings indicate that other physiological processes, beyond a rise in JH alone, are required for oviposition and colony initiation.

## 1. Introduction

Juvenile hormone (JH) has garnered significant attention from entomologists due to its multifaceted effects, ranging from orchestrating metamorphosis alongside ecdysteroid hormones, regulating female fertility by stimulating vitellogenin synthesis in the fat body and oocyte development, to generating complex polymorphisms in aphids and social insects (Hartfelder, 2000; Nijhout, 1998). In most insect species and in primitively eusocial Hymenoptera, such as some wasp species and bumble bees, JH-III acts as a gonadotropin, activating ovarian development and oogenesis (Röseler, 1977; Bloch et al., 1996; Bloch et al., 2000; Tibbetts and Sheenan, 2012; Kelstrup et al., 2014; Shpigler et al., 2016). By contrast, in some highly eusocial insect species like honey bees, some ant species, and termites, the role of JH has shifted from a gonadotropic one to being a regulator of maturation and division of labour within the colony (Hartfelder, 2000; Amsalem et al., 2014). An explanation for the transition in function can be found in the immunosuppressive effect of juvenile hormone (Rolff and Siva-Jothy, 2002; Pamminger et al., 2016) and the fact that highly eusocial insect species generally have long-lived queens (Keller and Genoud, 1997. Page and Peng, 2001). A proposed hypothesis posits that these species might have sidestepped the trade-off between reproduction and immunocompetence by unlinking JH from fecundity. This decoupling could enable reproduction under lower JH levels, consequently evading the immunosuppressive effects associated with heightened JH levels (Pamminger et al., 2016).

In bumble bees (*Bombus* spp.), JH levels fluctuate according to specific life cycle phases. Bumble bees in temperate climate zones have an annual lifecycle. A lone bumble bee queen founds a colony in spring, which will rapidly expand in size and worker population. In late

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summer, sexuals (males and new queens) are produced before colony demise. The young queens mate and then enter a phase of reproductive diapause during winter – also referred to as hibernation – until the following spring when they restart the cycle (Alford, 1975). Bumble bee queen hibernation is a physiological response to unfavourable conditions, characterised by minimal metabolic activity and a suspension of reproductive processes (Amsalem et al., 2015). Prior to hibernation, young mated queens exhibit low JH levels and underdeveloped ovaries (Chen et al., 2021; Röseler and Röseler, 1996, 1998). Prior to hibernation, young queens will build up fat body, which is their main energy reservoir during hibernation (Arrese and Soulages, 2010). During hibernation, JH levels remain low, which coincides with an arrested ovary development, It is only several days after exiting diapause that the corpora allata is reactivated and synthesis of JH increases, leading to elevated circulating JH levels in the haemolymph (Larrere et al., 1993). This rise in JH levels in the haemolymph is concurrent with ovary development (Chen et al., 2021; Röseler and Röseler, 1996, 1998). Egg production in the ovaries requires the allocation of lipid reserves for protein synthesis, primarily vitellogenin, a key yolk protein, which is transported to the developing eggs (Hartfelder, 2000). JH induces an upregulation of many fat body genes, activating the secretion of vitellogenin from the fat body into the haemolymph where it is then transported to the developing oocytes (Postlethwait and Giorgi, 1985; Riddiford, 2012; Shpigler et al., 2014, Santos et al., 2019; Shpigler et al., 2020).

Other known physiological processes in which JH plays a key role in bumble bees include metabolism (Shpigler et al., 2021), caste determination (Cnaani et al., 2000; Bortolotti et al., 2001), wax production, and the production of the honey pots (Shpigler et al., 2014). For bumble bees, JH application has been demonstrated to stimulate ovarian development of workers, vitellogenin production, as well as dominance behaviour (Röseler, 1977; van Doorn, 1989; Shpigler et al., 2010), although another study did not find a link between JH and vitellogenin synthesis (Amsalem et al., 2014).

The multi-faceted effects of juvenile hormone are often evaluated by using juvenile hormone analogues. The permeability of the insect cuticle for JH, in combination with the fact that high dosages can completely disrupt an insect's metamorphosis, gave rise to the production of JH analogues as insecticides (Nijhout, 1998). Methoprene is a juvenile hormone analogue with similar behavioural and physiological effects as JH (Robinson and Vargo, 1997; Giray et al., 2005), including its gonadotropic effects in most insects (Riddiford, 2012). It is hypothesized that methoprene binds directly to the JH receptors but does not affect the circulating juvenile hormone levels in the haemolymph, causing direct effects on physiological or behavioural processes regulated by JH (Huang et al., 2018).

The effects of treatment with JH or its analogues on oviposition and colony initiation in bumble bee queens are, to our knowledge, not assessed thus far. In this study, we evaluate how different factors, such as hibernation state, caste, and species, influence the gonadotropic effect of juvenile hormone in bumble bee queens using topical application of different doses of the JH analogue methoprene. Previous studies have focused on determining JH effects on ovary development as a sign of reproduction without directly evaluating egg laying and colony initiation. Here, we assess whether exogenous methoprene administration stimulates or accelerates ovary development and subsequent colony initiation in hibernated and non-hibernated queens of the two model bumble bee species *Bombus terrestris* and *Bombus impatiens*. In addition, the effect of methoprene treatment on ovary development in callow workers of both species is assessed with the aim to compare sensitivity of the worker caste of both species to the gonadotropic effect of juvenile hormone.

## 2. Materials and methods

### 2.1. Gonadotropic effects of methoprene in hibernated and non-hibernated queens

We studied two commercially available bumble bee species, namely *Bombus terrestris* (Linnaeus) and *Bombus impatiens* (Cresson) (Biobest Group NV, Westerlo, Belgium). All queens used for these experiments were mated at the age of seven days. For each species, two groups of queens were selected: in the first group, one-week-old mated queens were promptly treated after mating (<two hours between mating and treatment); for the second group, one-week-old mated queens were artificially hibernated for 17 weeks at 4 °C and 85% relative humidity in a large cooling chamber. A hibernation time of 17 weeks was chosen to approach the natural hibernation time of *B. terrestris,* which is typically 6–9 months (Alford, 1969) and to approach the optimal hibernation time under artificial conditions (Beekman et al., 1998). The queens in the second group were treated promptly after being taken from the cooling chamber, while they were not fully active yet. Each group of queens originated from a mixture of parental colonies. For each species and hibernation condition, queens were randomly divided over different topical treatments. Each treatment group consisted of 10 queens for *B. terrestris* or 7–8 queens for *B. impatiens* (8 for the non-hibernation condition, 7 for the hibernation condition). Treatments consisted of an acetone control and different doses of methoprene: 75, 150, 300, and 600 μg/queen for *B. terrestris*, and 100, 500, and 1000 μg/queen for *B. impatiens*. The discrepancy in doses used for both species stems from sample size limitations for *B. impatiens*. Concentrations were based on previous studies on Hymenoptera which used single topical doses of 20–250 μg methoprene per individual (Agrahari and Gadagkar, 2004; Giray et al., 2005; Oliveira et al., 2017; Bortolloti et al., 2020). We chose our dose range to exceed those from the studies mentioned above to account for the larger body size of bumble bee queens compared to bumble bee workers or wasp queens that were used in the aforementioned studies. Five μL of each treatment solution containing different concentrations of methoprene (Merck, Darmstadt, Germany; 0, 15, 30, 60 and 120 μg/μl for *B. terrestris*, and 0, 20, 100 and 200 μg/μl for *B. impatiens*, respectively) in acetone – or acetone only in case of the control treatment – was pipetted onto the dorsal side of the queens' abdomens using a micropipette. After application of the solution, every queen was held between forceps for a duration of 30 s to ensure absorption of the methoprene and was then individually placed in a plastic rearing container (16 × 16 × 10.5 cm). Queens were kept in a climate-controlled room at 28 °C and a relative humidity of 55%, in complete darkness except during observations. All queens were fed *ad libitum* 50% Brix sugar solution (Biogluc®, Belgosuc, Belgium) and a mix of gamma-irradiated honeybee-collected pollen which were mixed with a 15% sugar solution (72% Biogluc®) and kneaded into pollen balls. Mortality and oviposition were checked daily for a total of five days for *B terrestris* and six days for *B. impatiens,* when the first oviposition can be expected based on previous observations under the same climatic conditions (unpublished data). At the end of each experiment, the queens were collected and frozen at −20℃ for dissection. Dissections of the queens were performed in order to assess the level of ovarian development according to the levels described by Duchateau and Velthuis (1989), whereby 0 corresponds to ovaries in which oocytes and trophocytes are not visible, I to ovaries containing spherical small oocytes and longer trophocytes, II to ovaries containing rounded oocytes and longer trophocytes, III to oval-shaped oocytes that are larger than the trophocytes, IV to ovaries containing at least one mature oocyte with chorion and degenerated trophocytes, and $IV^R$ corresponding to ovaries with signs of oocyte resorption. For further binomial analysis, ovaries were considered developed when they reached developmental stage III or higher, which corresponds to ovaries having at least one oocyte larger than the trophocyte follicle. We included ovaries with signs of resorption as developed because oocyte resorption is common in egg-laying queens

(Duchateau and Velthuis, 1989).

### 2.2. Gonadotropic effects of methoprene in callow workers

For both *B. terrestris* and *B. impatiens*, callow workers were collected and divided into five different groups: four methoprene treatment groups and one control group of pure acetone. Lower concentrations of methoprene, diluted in acetone, were used to account for the smaller body size and mass of the workers compared to the queens. The methoprene concentrations applied were 20, 40, 80, and 160 μg/worker, which was approximately 4 times lower than doses applied to queens, as a worker's body mass is on average 1/4th lower compared to that of queens. As described for queens, 5 μl of a methoprene solution in acetone was applied to the abdomen of each worker (0, 4, 8, 16 and 32 μg/μl solution, respectively). Sample sizes for each group included 10 workers for *B. terrestris* and 7 workers for *B. impatiens*, respectively. Workers were kept individually in plastic containers as described for queens in section 2.1. The experiment again lasted for 5 or 6 days (for *B. terrestris* and *B. impatiens*, respectively), during which oviposition and mortality was assessed on a daily basis. Finally, workers were collected and frozen for dissection as described in section 3.1.1. In addition, thorax widths were measured using digital callipers (Mitutoyo, Kanagawa, Japan). Thorax width is a standard measurement for body size in bumble bees (Goulson, 2003).

### 2.3. Long-term effects of methoprene on colony development

A total of 160 *B. terrestris* queens with 20 queens per treatment group was used to assess long-term effects of methoprene on oviposition and colony development. There were two hibernation conditions: queens were either hibernated for a 17-week-period or collected right after mating without hibernation. For each hibernation condition there were four methoprene treatments: an untreated control, an acetone control of 5 μl pipetted once onto the abdomen, a single application of 100 μg methoprene per queen (5 μl of a 20 μg/μl methoprene solution in acetone pipetted onto the abdomen), and a single application of 500 μg methoprene per queen (5 μl of a 100 μg/μl methoprene solution in acetone pipetted onto the abdomen). For *B. impatiens*, there was only the hibernation condition due to rearing population size limitations, for which 80 20-week-hibernated queens were divided over the four treatments ($N = 20$ per treatment). These hibernated *B. impatiens* queens were divided over the same 4 treatment groups as described for *B. terrestris* queens. For both study species, queens were placed in rearing containers as described in section 2.1 after the single treatment (or no treatment in case of the untreated controls), with *ad libitum* access to 50 °Bx sugar water and *ad libitum* access to pollen. Rearing boxes were kept in the dark (except during feeding/observations), at an ambient temperature of 28 °C and 50–55% relative humidity. The number of days until oviposition, queen mortality before oviposition, colony foundation success (i.e. whether at least one worker was produced) and number of days until the first worker emerged were assessed. The experiment concluded on day 56 for *B. terrestris* and day 63 for *B. impatiens*, reflecting the species-specific developmental time discrepancy. This duration corresponded to the time required for 50% of hibernated queens to attain a colony size of 100 workers. For non-hibernated *B. terrestris* queens, the experiment also terminated after 63 days, aligning with the hibernated counterparts.

### 2.4. Statistical analysis

The data supporting the findings of this study are available in the supplementary material of this article. Model selection was performed based on the Akaike Information Criterion (AIC), and model assumptions were tested using the package DHARMa (Hartig, 2020). All statistical analyses were performed in R 4.2.0 (R core team, 2022) and the script used to run the analyses is included as supplementary material.

The effect of exogenous methoprene application on queen mortality was assessed by using a general linear model (GLM, R-package *stats*) with binomial distribution ($1 = $ dead, $0 = $ alive) and methoprene dose and hibernation status as fixed factors. For the analysis of methoprene effects on queen ovary development, general linear models with binomial distribution were used. Ovary development was a binomial variable with developmental stages 0, I and II being categorised as 'undeveloped' and developmental stages III, IV and $IV^R$ as 'developed'. In the final model, methoprene dose was included as a continuous predictor, along with fixed factors hibernation status and species, the interaction effect between dose and hibernation status, and the interaction effect between methoprene dose and species. To assess whether methoprene treatment caused increased mortality in callow workers, a binomial GLM was used with methoprene dose as a continuous predictor and species as a fixed factor. For the effect of methoprene on ovary development of callow workers, a similar binomial GLM was used as for mortality, yet also including body size as a covariate. Interaction effects were not included because they significantly increased the AIC of the model.

To assess long-term effects of methoprene administration on colony development, binomial GLMs were performed to assess the effect of methoprene treatment on queen mortality and colony foundation success as binomially distributed dependent variables. Treatment was considered as a categorical predictor with four levels (untreated control, acetone control, 100 μg/queen and 500 μg/queen). Post-hoc comparisons were conducted using *contrast* function, with "trt.vs.ctrl" and Tukey-adjustment in the package *emmeans*. The number of days until egg laying was assessed using a survival analysis, with the trial termination date as cut-off point (R-packages *survival*, Therneau, 2020, and *survminer,* Kassambara and Kosinki, 2018). Queens that died shortly after start of the trial (before oviposition occurred) were excluded from the analysis for time to oviposition but were included for the analysis of colony foundation success, as death was considered as a failure to start a colony.

## 3. Results

### 3.1. Gonadotropic effect of methoprene

Hibernation status had a significant effect on queen mortality for both species (likelihood ratio $\chi^2$ test (further referred to as $\chi^2$) $= 5.697$, $p = 0.017$), regardless of methoprene treatment ($\chi^2 = 0.749$, $p = 0.387$; no significant interaction effects $p > 0.05$; Fig. S1). Queen mortality was low in non-hibernated queens (0% in *B. impatiens* queens and 1.7% in *B. terrestris* queens), but higher in hibernated queens (5.0% mortality for hibernated *B. terrestris* queens and 14.3% mortality in hibernated *B. impatiens* queens). Hibernation status affected the ovarian development in both bumble bee species (main effect of hibernation: $\chi^2 = 59.773$, $p < 0.001$; no significant interaction between hibernation and species $p > 0.05$; Fig. 1). For both species, the proportion of queens with developed ovaries by the end of the test period was significantly higher for hibernated queens (100% for *B. terrestris* and 90% *for B. impatiens*, respectively) compared to non-hibernated queens (54.2% for *B. terrestris* and 12.5% for *B. impatiens*). Further, there was a significant interaction between hibernation status and methoprene dose (interaction effect: $\chi^2 = 4.046$, $p = 0.044$), indicating that methoprene only affected ovary development in a dose-dependent matter when queens were not hibernated (Fig. 1). Hibernated queens for both species mostly had developed ovaries, regardless of methoprene concentration, while the proportion of queens with developed ovaries increased with methoprene administration when queens of both species were not hibernated. There was a significant interaction between methoprene dose and species ($\chi^2 = 15.374$, $p < 0.001$), indicating that there was a stronger dose-dependent effect for *B. terrestris* than for *B. impatiens* (Fig. 1). For non-hibernated *B. terrestris* queens, the lowest methoprene dose already resulted in a substantial increase in the proportion of queens with developed ovaries

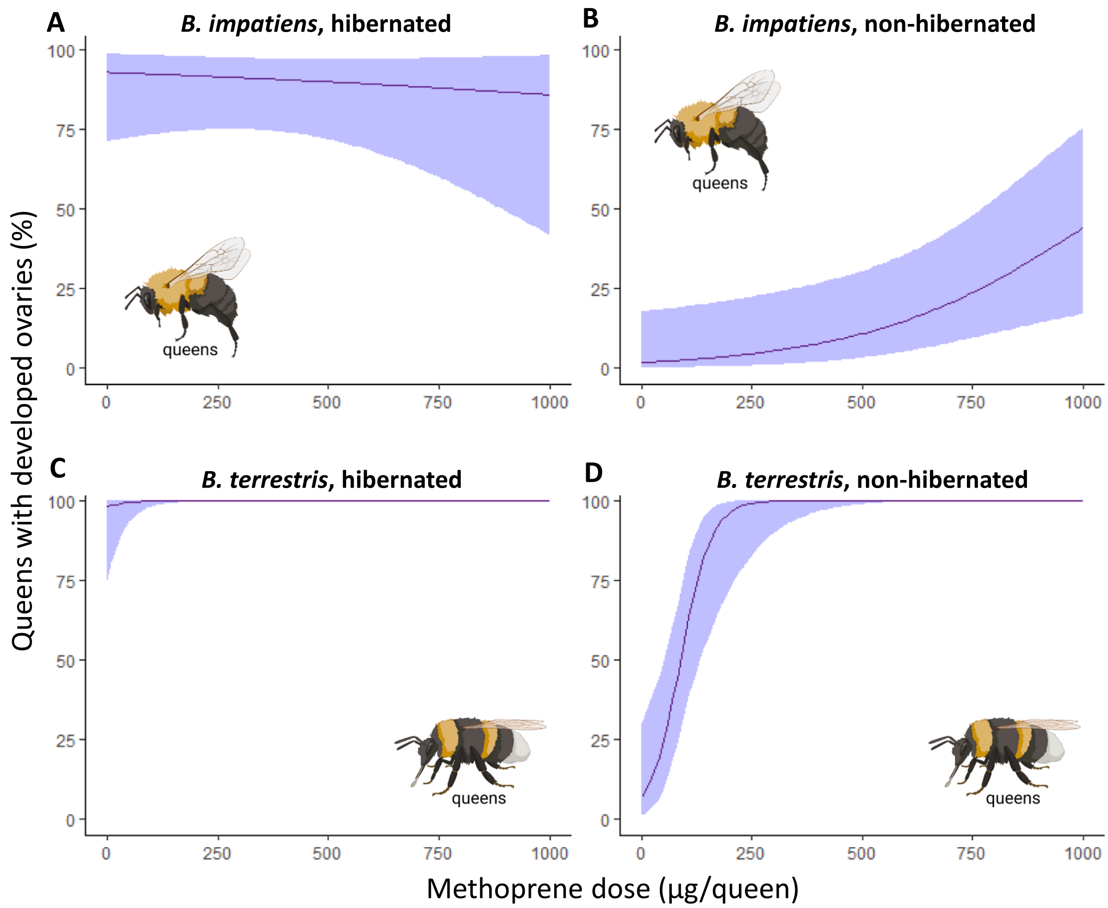


**Fig. 1.** Effect of the dose of a single methoprene treatment on the proportion of isolated queens with developed ovaries after 5–6 days for **(A)** hibernated *B. impatiens* queens (N = 28), **(B)** non-hibernated *B. impatiens* queens (N = 32), **(C)** hibernated *B. terrestris* queens (N = 50) and **(D)** non-hibernated *B. terrestris* queens (N = 50). Curves represent the prediction of the statistical model with 95% confidence intervals.

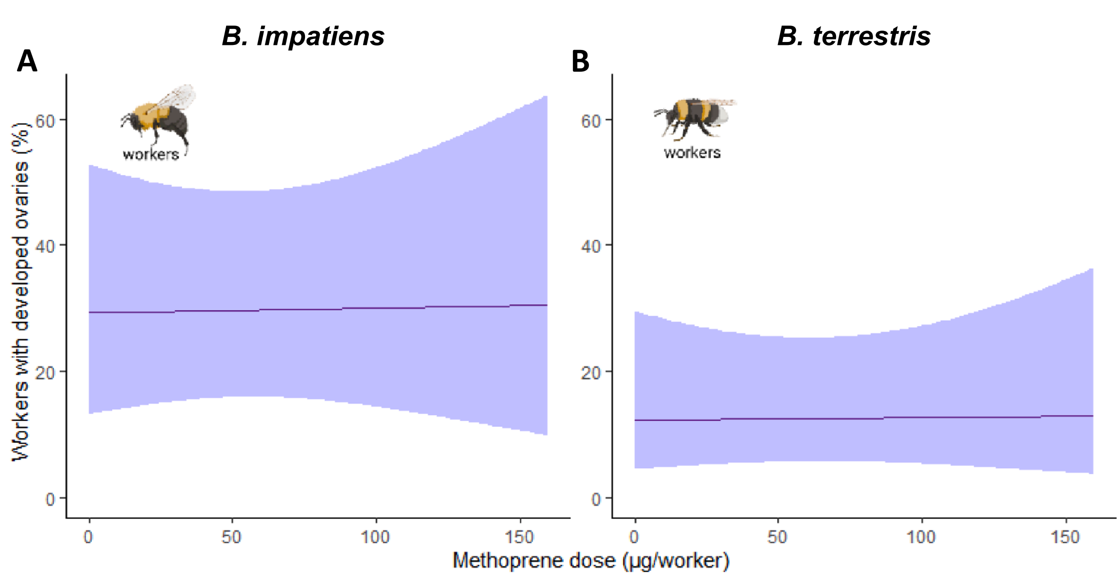


**Fig. 2.** Effect of the dose of a single methoprene treatment on the proportion of isolated callow workers with developed ovaries after five or six days for **(A)** *B. impatiens* workers (N = 35) and **(B)** *B. terrestris* workers (N = 50). Curves represent the prediction of the statistical model with 95% confidence intervals.

(50% of queens with developed ovaries for 75 μg/queen compared to 0% for the control treatment) and 100% of queens had developed ovaries when received a dose of 300 μg or higher. For non-hibernated *B. impatiens,* however, 0% of the control queens had developed ovaries and the highest dose of 1000 μg/queen was not sufficient to provoke ovary development in all queens, as only 37.5% of queens had developed ovaries (Fig. 1.).

In callow workers, overall mortality was very low (only three dead workers over all experiments), and there was no significant effect of methoprene dose on ovary development ($\chi^2 = 0.248$, p = 0.619). We observed a trend for a higher proportion of ovary development for *B. impatiens* workers compared to *B. terrestris* workers (27.5% compared to 15.3%; main effect species: $\chi^2 = 3.293$, p = 0.070), but methoprene did not affect ovary development for either species ($\chi^2 = 0.005$, p = 0.946; Fig. 2A). However, the probability for developed ovaries increased significantly with thorax width as a measure for body size for both species ($\chi^2 = 5.461$, p = 0.019; Fig. S2).

### 3.2. Effect of exogenous methoprene administration on colony foundation

In the long-term experiment, queen mortality was low overall for *B. terrestris* queens (6% for non-hibernated queens and 1% in hibernated queens) and methoprene treatment did not affect queen mortality (hibernated queens $\chi^2 = 2.811$, p = 0.422; non-hibernated queens $\chi^2 = 6.914$, p = 0.074). Colony foundation success (i.e. at least one worker born) was much higher for hibernated *B. terrestris queens* (97%) by the end of the experiment compared to non-hibernated queens (4%), but again, no significant effect of methoprene treatment was observed (hibernated queens $\chi^2 = 1.765$, p = 0.623; non-hibernated queens $\chi^2 = 4.643$, p = 0.120). By contrast, hibernated *B. impatiens* queen mortality was higher for the 500 μg/queen methoprene dose (45%) compared to the other treatments (5%, 15% and 5% for the untreated control, acetone control and 100 μg/queen, respectively), with mortality for the 500 μg/queen dose being significantly higher compared to the untreated control (post-hocz = 2.450, p = 0.042; Fig. 3A). Colony foundation success was lower for the highest methoprene dose as well (55% for the 500 μg/queen concentration compared to 90% for the untreated control, 85% for the acetone control and 90% for the 100 μg/queen treatment, respectively), with a marginally significant difference between the highest dose and the untreated control (post-hoc z = -2.294, p = 0.064; Fig. 3B). Hibernated queens oviposited after 8 days and 14 days on average for *B. terrestris* and *B. impatiens*, respectively, while non-hibernated *B. terrestris* queens oviposited much later (after 48 days on average), which resulted in a delay in colony development of approximately 40 days for non-hibernated *B. terrestris* queens. Methoprene application had no effect on the timing of oviposition for either species regardless of hibernation status (*B. terrestris* hibernated $\chi^2 = 0.2$, p = 1.0; *B. terrestris* non-hibernated $\chi^2 = 2.8$, p = 0.4; *B. impatiens* hibernated $\chi^2 = 3.6$, p = 0.3; Fig. 4).

## 4. Discussion

The gonadotropic effect of juvenile hormone (JH) is extensively documented in insects (Robinson and Vargo, 1997; Hartfelder, 2000; Santos et al., 2019), although complex interactions with social environment and caste can emerge in social insects. While JH retains its gonadotropic function in many primitively eusocial species (e.g. bumble bees: Shpigler et al., 2020; Amsalem et al., 2014 and wasps: Agrahari and Gadagkar, 2003; Tibbetts and Sheenan, 2012), this is not universally observed in highly eusocial insects (Robinson and Vargo, 1997). This study aims to investigate factors influencing the JH-induced gonadotropic effect, encompassing queen hibernation state, caste, and species by utilizing two model bumble bee species, *B. terrestris* and *B. impatiens*. Additionally, our investigation extends to evaluating the direct impact of JH on queen oviposition, transcending evaluation of merely ovarian development and oogenesis. Our results demonstrate that a single exogenous application of the JH analogue methoprene affects ovary development in bumble bees differently according to caste, species and hibernation status. Yet, despite the observed gonadotropic effect in queens, oviposition itself was not induced by methoprene treatment.

Hibernated queens typically activate their ovaries shortly after hibernation (Chen et al., 2021; Röseler and Röseler, 1986; Röseler and Röseler, 1988), which was confirmed in our study. JH levels are typically low before and during hibernation but increase after breaking hibernation, coinciding with ovary development (Chen et al., 2021). As a consequence, gonadotropic effects of exogenous JH are expected when queens are not hibernated, as JH levels are naturally low then. Our results support this hypothesis, as methoprene stimulated ovarian development in non-hibernated queens while it did not affect ovarian development in hibernated queens, presumably due to naturally high JH levels in hibernated queens. A putative explanation for these observed differences between hibernated and non-hibernated queens therefore lies in the differences between both the natural JH levels before and after hibernation. The gonadotropic effect of JH is attributed to its stimulatory effect on the production of vitellogenin in the fat body, which in turn stimulates ovary development and oogenesis (Shpigler et al., 2014; Santos et al., 2019).

The effect of methoprene in queens was more pronounced in *Bombus terrestris* than in *Bombus impatiens*, with 100% of *B. terrestris* queens having developed ovaries after administration of 300 μg/queen, while only 37.5 % of *B. impatiens* queens had developed ovaries at the highest tested dose (i.e. 1000 μg/queen) within five and six days after mating, respectively. These results can indicate a lower sensitivity of *B. impatiens* to the exogenous JH analogue methoprene or to JH. It remains to be investigated whether natural JH levels differ between both species, as this could explain the observed differences between the species.

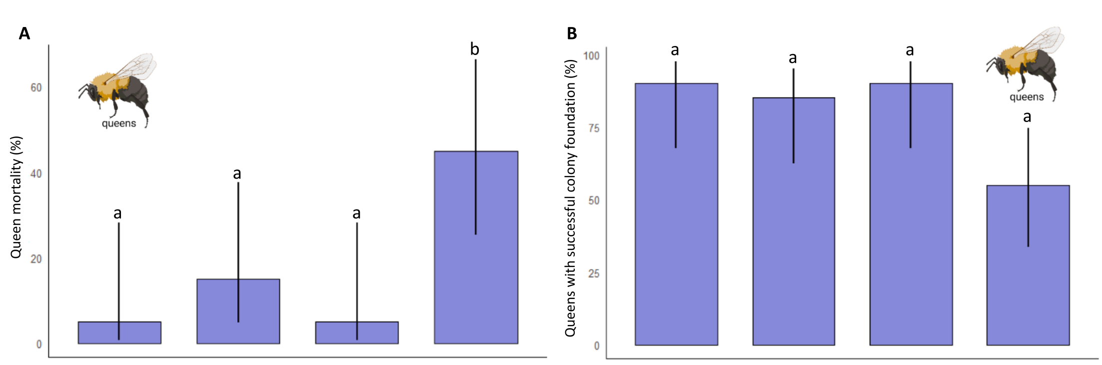


**Fig. 3. (A)** Proportion of *B. impatiens* queens (N = 80) of the long-term experiment that died before oviposition. **(B)** Proportion of *B. impatiens* queens (N = 80) that successfully started a colony (i.e. produced at least one adult worker). Bars represent the estimated marginal means of the statistical model including 95% confidence intervals. Letters indicate significant differences of treatment groups to the untreated control.

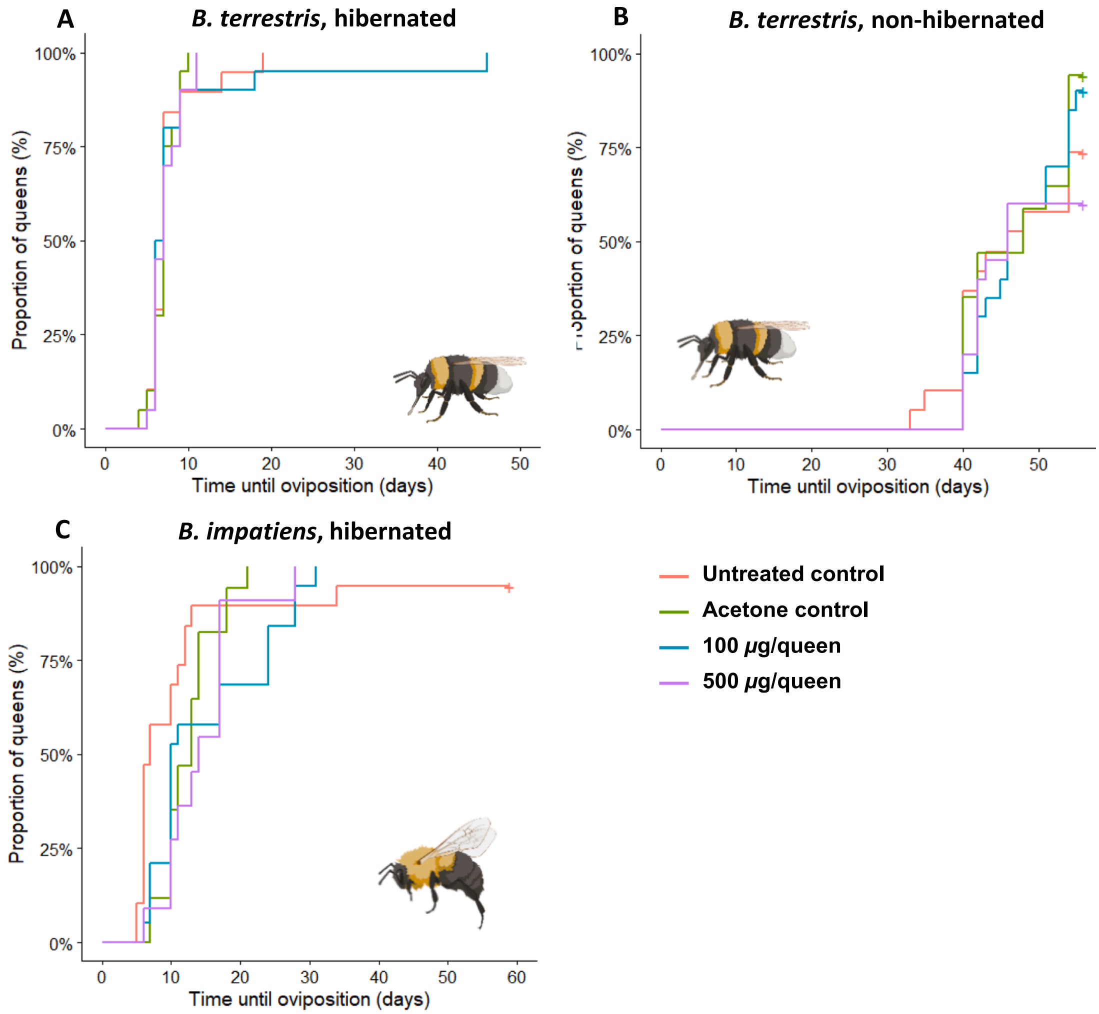


**Fig. 4.** Proportion of queens that oviposited over time for different control and methoprene treatments in **(A)** hibernated *B. terrestris* queens (N = 79), **(B)** non-hibernated *B. terrestris* queens (N = 76), and **(C)** hibernated *B. impatiens* queens (N = 66). Queens that died prior to oviposition were excluded from analysis.

For both study species, there was no gonadotropic effect of exogenous JH in workers at the tested doses. Yet, induction of ovary development by JH application has previously been demonstrated in bumble bee workers. For instance, a topical treatment of JH-III (70 µg/bee dissolved in dimethylformamide (DMF)) stimulated ovary development in queenright bumble bee workers (Shpigler et al., 2010). Additionally, workers with removed corpora allata (allatectomy) which consequently do not produce JH, did not develop their oocytes at all. However, a single or double topical application of JH-III has been shown to (at least partially) reverse the negative effects on oocyte development in allatectomized bumble bee workers (Shpigler et al., 2014). Similarly, Röseler (1977) induced ovary development in callow *B. terrestris* workers by injecting them with JH-I. Discrepancies between the studies mentioned above highlight the importance of application method (e.g. injection vs. topical application; acetone or DMF as carrier for topical application) to compare results between studies. Our concentration and those of the aforementioned studies might not have been sufficient to elicit gonadotropic effects in workers through topical application. In addition to dosage of a single application, repetition of applications may also impact the gonadotropic effects of JH or its analogues (Shpigler et al., 2014). Another critical factor in the activation of worker ovarian development is the social environment. Most studies mentioned above kept workers in groups during the experiment (e.g. Röseler, 1977; Amsalem et al., 2014; Shpigler et al., 2014; Bortolloti et al., 2020). Outcomes of studies on reproductive state likely depend on whether workers are kept alone, in queenless groups or queenright colonies (Bloch et al. 2000; Amsalem et al., 2014). In the current study, workers were kept alone, which excludes any possible effects of social interactions. We hypothesize that social interactions have an additive effect on the JH pathway for the initiation of reproduction. We support this hypothesis by the finding that allatectomized workers did not develop ovaries, even in groups (Shpigler et al., 2014), suggesting that the social environment has no influence on reproduction in the absence of JH. On the other hand, the effect of JH on queens' ovary development is enhanced in groups vs. isolated queens (Shpigler et al., 2021). In further support of this hypothesis, the presence of workers strongly affects the queen oviposition rate (Woodard et al., 2013). Nevertheless, queens do not require social interaction for colony initiation, which indicates that other signals besides the social environment are involved. In nature, the queens collect food, find a nest site and build honey pots before laying eggs. The effect of the social environment and the nest establishment may be mediated by similar chemicals such as biogenic amines like dopamine or through other allatoregulatory neuropeptides (Verlinden et al. 2013; Verlinden et al. 2015; Brent et al., 2015).

Exogenous methoprene administration did not affect timing of first egg laying in both *B. impatiens* and *B. terrestris* queens. In *B. terrestris*, hibernated queens laid eggs 40 days sooner on average compared to non-hibernated queens, regardless of methoprene treatment. The precise physiological mechanisms underpinning the acceleration of egg laying with extended hibernation remain elusive. However, prior investigations have drawn a correlation between shorter pre-oviposition periods and the extended hibernation periods in bumble bee queens (Beekman et al., 1998; Gosterit and Gurel, 2009). Exogenous methoprene application, however, could not accelerate egg laying in non-hibernated queens, which is highly remarkable because of its demonstrated gonadotroptic effect. Under the assumption of a similar mode of action between JH and methoprene (Ashok et al., 1998; Parthasarathy and Palli, 2009), this finding indicates that ovary development and egg laying are regulated by physiological pathways that are not directly linked to each other. This suggestion was previously made by Robinson and Vargo (1997) based on earlier findings of workers with mature oocytes found in queenright colonies that would not lay eggs in the presence of the mother queen (Duchateau and Velthuis, 1989) and the observation that queens are able to lay eggs despite low rates of JH biosynthesis (Larrere and Couillaud, 1993). Amsalem et al. (2014) noted an uncoupling of JH and vitellogenin in *B. terrestris*, implying the regulation of vitellogenin by factors other than JH. This suggests that vitellogenin and oviposition might be influenced by alternate physiological pathways, independent of JH. A possible route is that nutrient scarcity during diapause impacts nutrient-sensitive pathways like the target of rapamycin (TOR) and insulin/insulin-like growth factor signaling (IIS), which can directly or indirectly trigger vitellogenesis through endocrine pathways (reviewed in Kapheim, 2017). This can provide an explanation for our findings, whereby diapause accelerated queen egg laying independent of JH treatment and JH failed to induce oviposition in non-hibernated queens despite its effect on ovary development.

In the long-term experiment, the highest concentration of 500 µg methoprene per queen led to an increased mortality before oviposition in *B. impatiens* queens, which led to a lower colony foundation success. The high mortality could reflect toxicity of the chemical at high concentrations or it could indicate that juvenile hormone plays an important role in bumble bee adult lifespan. In many other insects, including the model insect species *Drosophila melanogaster*, it has been shown that juvenile hormone is a pro-aging hormone that reduces adult lifespan (Herman and Tatar, 2001; Hodkova, 2008; Yamamoto et al., 2013). In line with the hypothesis that JH has pro-aging effects is the fact that JH levels are typically low during reproductive diapause, a period during which there is an increased life span and increased stress resistance (Amsalem et al., 2015). Moreover, in species characterized by a highly eusocial structure and extended queen lifespans, JH has relinquished its gonadotropic role, which allows queens to reproduce under low JH levels and to circumvent the immunosuppressive repercussions of elevated JH levels (Pamminger et al. 2016).

In conclusion, our results demonstrate that the effects of JH on bumble bees vary according to species, castes and hibernation status. For both bumble bee species, there were clear gonadotropic effects in non-hibernated queens and no detectable gonadotropic effects in workers. Hibernation increased ovary development and shortened time to egg laying, likely due to a natural rise in JH levels as a result of breaking hibernation. Exogenous methoprene application could not shorten time to egg laying or increase colony foundation success, despite the demonstrated gonadotropic effect in non-hibernated queens. This indicates that JH may not play a direct role in the physiological process of egg laying. Further research is required to clarify and disentangle the physiological processes required for egg laying in bumble bees.

## 5. Author statement

Conception or design of the work: EW, AVO.
Data collection: CVD, NL, ET.
Data analysis and interpretation: EW, CVD, SP, HS, TW, AVO.
Drafting the article: EW, CVD, SP, AVO.
Critical revision of the article: EW, SP, HS, AVO.
Final approval of the version to be published: all authors.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

data and code are provided as supplementary data

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jinsphys.2023.104557.

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